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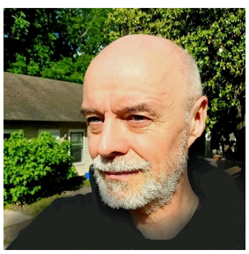
AUGUSTANA UNIVERSITY
COSC 2230 Computer Science III
Fall Term 2026
Where: FSC 372 | **When:** MWF 1:00-1:50 pm
Canvas: https://augie.instructure.com/courses/15648
AI use: permissive (details below)



Instructor

Office/student hours: in room FSC 386 - book via Navigate
Tue 1:00 – 3:00 pm, Thu 9:00 am – 12:00 pm
Phone: 605.274.4285 | Email: marcus.birkenkrahe@augie.edu
More information: birkenkrahe.github.io or linkedin.com/in/birkenkrahe

Course Description

This course is a continuation of Computer Science II and provides a deeper look into object oriented programming, generic types, lambda functions, stream processing, and concurrent processing. It also presents several basic data structures: stacks, queues, lists, trees, and graphs. Algorithms for sorting, searching, and memory management are also introduced.

Course AI Classification

AI Use in this Coursework: *Permissive*.
Students may use AI tools freely throughout this course, including on assessments, unless otherwise specified, provided use is disclosed and outputs are critically evaluated. AI tools are not permitted during exams unless explicitly stated on the exam itself. See Augustana University AI Policy.

Course Outcomes

Students who complete this course will be able to:

- Design, implement, and test encapsulated reference types with a well-formed API, including immutable types and wrapper classes (autoboxing/unboxing)
- Understand and apply interface-based polymorphism (dynamic dispatch) to write code that works against a contract, not an implementation
- Evaluate the time and space efficiency of a program using order-of-growth analysis and empirical benchmarking
- Understand and implement fundamental sorting and searching algorithms, and compare their performance
- Design and use stack and queue data structures (array-based and linked-list-based), including generic, iterable client code via Java's Iterable/Iterator mechanism
- Understand and use symbol tables (hash tables and binary search trees) to associate keys with values
- Model networks as graphs, implement graph-processing clients, and analyze real-world graphs for properties such as the small-world phenomenon (shortest paths)

University Policies and Resources

The following *link* explains University-wide academic policies, including Accessibility & Accommodations, Canvas, Honor Code, Illness Prevention & Protection, Student Well-being/Mental Health, Title IX Statement and other Campus Resources (including Help Desk, Registrar's Office, & Writing Center).

Course Policies and Expectations (optional policies noted)

Attendance/Punctuality

This is a face-to-face course and you need to attend class to be successful in this topic. Many of our assignments will be started as in-class activities and will require your presence. If you are absent, make arrangements with a classmate to get class notes and a briefing on discussion topics, watch session recordings if available, and review the online material. Classes will start at the scheduled time and you should arrive 2-3 minutes prior to the start of class.

Generative Artificial Intelligence Policy

AI tools predict, they don't think, and they will confidently hand you wrong answers alongside right ones. Learning to verify, correct, and take responsibility for AI-assisted work — propose, test, find where it breaks, revise — is part of what this course teaches, and it's also just good scientific practice. What isn't allowed is using AI as a substitute for that judgment: undisclosed use, or submitting output you haven't checked, defeats the purpose. For more details on "should you use AI to help you code", see here: https://github.com/birkenkrahe/org/blob/master/fall24/UsingAItoCode.pdf

Course Modification (inclement weather, illness, etc.) Policy

If significant inclement weather, professor illness, or any special circumstance arises that may require an adjustment to the course schedule (i.e. an alternative assignment, switching to synchronous online, canceling the class, etc.), every attempt will be made to communicate such changes to students by the night before the scheduled class. Communication to students will come via an announcement on the class LMS and/or an email to your AU email address. If you have reason to believe the class may not be able to be offered as scheduled, please keep an eye on these inboxes.

Exam Policy

Exams during the term are based on material discussed in class. Make-up exams will only be given if a written excuse (a physician's note or evidence of a University- excused absence) is submitted to me prior to the exam. Make-up exams may be different than in-class exams and the material may generally be more difficult. During exams, no cell phones, calculators, or other electronic devices may be used. All notes and books must be out of sight, except for materials allowed (as specified during the review session). You may not leave the room during the exam.

Course Materials and Access

Required textbook

Introduction to Programming in Java: An Interdisciplinary Approach, 2nd Edition
by Robert Sedgewick & Kevin Wayne, Addison-Wesley Professional, 2017 ISBN-13: 978-0672337840
Book Website: https://introcs.cs.princeton.edu/java/home/. The book has a site with excerpts and many other resources: https://introcs.cs.princeton.edu/java/home/ (with free excerpts)

Other optional books (legally free and online):

- Introduction to programming using Java, version 9 (2022) https://math.hws.edu/javanotes/
- Think Data Structures - Algorithms & Information Retrieval in Java, version 1, by Downey (2016) https://greenteapress.com/thinkdast/thinkdast.pdf
- Open data structures java edition - An open content textbook by Morin (2013), https://opendatastructures.org/
- An Open Guide to Data Structures and Algorithms by Bible/Moser (2023), https://pressbooks.palni.org/anopenguidetodatastructuresandalgorithms/

Course platforms

- GitHub for lectures, code: https://github.com/birkenkrahe/cosc_2230_fall_26
- Google Chat Space for discussion and sharing: https://tinyurl.com/gspace-2230
- Google Drive for whiteboard photos and papers: https://tinyurl.com/fa26-photos
- DataCamp for additional optional practice: https://www.datacamp.com
- YouTube Playlist: https://tinyurl.com/youtube-3430

Assignments and Grading Procedures

Grading table:

Letter Grade	Percentage Range
A	93–100%
A-	90–92%
B+	87–89%
B	83–86%
B-	80–82%
C+	77–79%
C	73–76%
C-	70–72%
D+	67–69%
D	63–66%
D-	60–62%
F	Below 60%

Grade breakdown

What	When	How much
Project (Scrum)	monthly	25%
Quizzes (not timed)	weekly	15%
Programming assignments	weekly	40%
Midterm exam (50 min)	TBD	10%

What	When	How much
Final exam (optional)	TBD	10%

- **Programming assignments:** at home, individual, AI allowed unless specified – use must be disclosed, outputs critically evaluated and tested. Estimated time: 40-55 hrs/term (3-4 hrs/week)
- **Term Project** (team of 2-3): 3 sprint reviews (3 x 5%) + presentation (10%)
- **Quizzes**: multiple choice, not timed, at home, individual
- **Midterm exam:** 50 min, individual, based on quizzes (with mock exam)
- **Final exam**: 30 min, optional, individual conversation

Course Schedule

One quiz and one programming assignment per week (see **Grading**).

Date	Topic	Notes
Wed (8/26)	Overview	Review the AI policy, clarify questions
Fri (8/28)	Setup/projects	
Mon (8/31)	Ch. 3 review 1/3	
Wed (9/2)	Ch. 3 review 2/3	Last day for online reg./adding w/o signature
Fri (9/4)	Ch. 3 review 3/3	
Mon (9/7)	NO CLASS - Labor Day	
Wed (9/9)	Ch.4.1 Performance	
Fri (9/11)	Ch.4.1 Performance	Independent Study/Internship paperwork due
Mon (9/14)	Ch.4.1 Performance	
Wed (9/16)	Ch.4.1 Performance	Last day to drop without signatures
Fri (9/18)	Ch.4.1 Performance	Extended chapel service (until 11:15a)
Mon (9/21)	Ch.4.2 Sorting & Searching	
Wed (9/23)	Ch.4.2 Sorting & Searching	

Date	Topic	Notes
Fri (9/25)	Ch.4.2 Sorting & Searching	Sprint 1: Project Proposal
Mon (9/28)	Ch.4.2 Sorting & Searching	
Wed (9/30)	Ch.4.3 Stacks & Queues	Graduation Application Deadline (10/1)
Fri (10/2)	Ch.4.3 Stacks & Queues	
Mon (10/5)	Ch.4.3 Stacks & Queues	
Wed (10/7)	Ch.4.3 Stacks & Queues	
Fri (10/9)	MIDTERM	
Mon (10/12)	NO CLASS - Native American Day	
Wed (10/14)	NO CLASS - Fall Break	
Fri (10/16)	Ch.4.3 Stacks & Queues	
Mon (10/19)	Ch.4.3 Stacks & Queues	
Wed (10/21)	Ch.4.3 Stacks & Queues	
Fri (10/23)	Ch.4.3 Stacks & Queues	Sprint 2: Literature Review
Mon (10/26)	Ch.4.4 Symbol Tables	
Wed (10/28)	Ch.4.4 Symbol Tables	Last day to drop with "W" or file S/U
Fri (10/30)	Ch.4.4 Symbol Tables	
Mon (11/2)	Ch.4.4 Symbol Tables	
Wed (11/4)	Ch.4.4 Symbol Tables	
Fri (11/6)	Ch.4.4 Symbol Tables	
Mon (11/9)	Ch.4.5 Case Study: Small World	
Wed (11/11)	NO CLASS - Veterans Day	
Fri (11/13)	Ch.4.5 Case Study: Small World	

Date	Topic	Notes
Mon (11/16)	Ch.4.5 Case Study: Small World	Interim 2027 & Spring 2027 registration (11/16-20)
Wed (11/18)	Ch.4.5 Case Study: Small World	
Fri (11/20)	Ch.4.5 Case Study: Small World	Sprint 3: Results / Abstract
Mon (11/23)	Ch.4.5 Case Study: Small World	
Wed (11/25)	NO CLASS - Thanksgiving Break	
Fri (11/27)	NO CLASS - Thanksgiving Break	
Mon (11/30)	Presentations	Sprint 4: Discussion
Wed (12/2)	Presentations	
Fri (12/4)	Presentations	Last day of classes
(12/7-12/11)	FINAL EXAM	Date/time/location TBD
		Final grades due Wed (12/16)